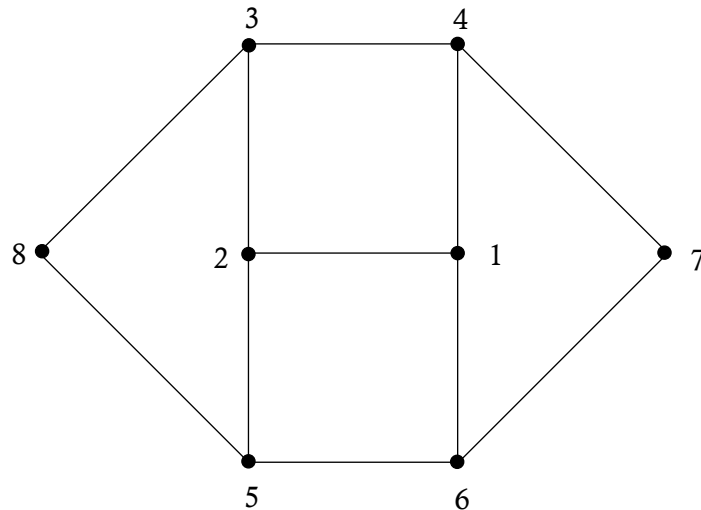


Graph Theory

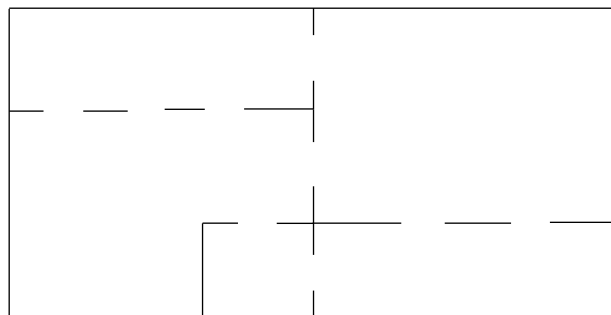
In a graph:

- The number of vertices is the **order** of the graph.
- The number of edges is the **size** of the graph.
- Two vertices are **adjacent** if there is an edge joining them
- The **degree** of a vertex is the number of edges incident at that vertex.
- A **walk** is any alternating sequence of vertices and edges that starts and finishes at a vertex.
- A **trail** is a walk that doesn't use any edge more than once.
- A **path** is a walk that doesn't visit any vertex more than once.
- An **Eulerian trail** is a trail that uses every edge in the graph.
- An **Eulerian circuit** is as an Eulerian trail that starts and finishes at the same vertex.
- A **Hamiltonian Path** is a path that visits every vertex in the graph.
- A **Hamiltonian cycle** is a walk that starts and finishes at the same vertex, and visits every other vertex exactly once.
- A graph is **connected** if there is at least one path between every pair of vertices.

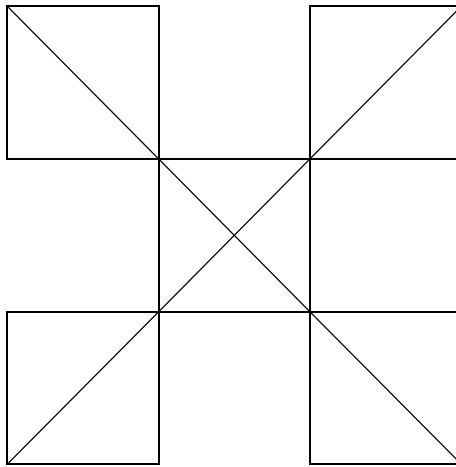
1. The graph shown below has set of vertices labelled 1, 2, ..., 8.



- Two vertices are adjacent in this graph if a simple mathematical property is satisfied by the two labels. Suggest what that property might be.
 - Use this graph to give an example of a trail that is not a path.
 - Find a Hamiltonian path in this graph.
 - Find a Hamiltonian cycle in the graph, or prove that none exists.
2. A simple graph contains vertices labelled B, E, I, N, O, S, V. In this graph two distinct vertices are adjacent if and only if their labels are consecutive in the word OBSESSIVENESS.
- Draw this graph.
 - Write down the degree sequence of this graph; what is its size?
 - Is it possible to draw this graph without taking your pen off the paper and without going over any edge twice? Explain your answer.
3. A multigraph contains vertices labelled B, E, I, N, O, S, V. For each pair of consecutive letters in the word OBSESSIVENESS put an edge between the (not necessarily distinct) vertices with those labels. This means there will be multiple edges between some vertices, and some vertices will have a loop.
- Draw this graph.
 - Write down the degree sequence of this graph; what is its size?
 - Is it possible to draw this graph without taking your pen off the paper and without going over any edge twice? Explain your answer.
4. The diagram below shows the floor plan of an office building. At the end of the day, the building manager needs to lock all of the doors (shown by gaps) before she leaves.
- Represent the floor plan as a graph, with the rooms as vertices and the doors as edges.
 - Show that there is a room in which the manager can start so as to pass through each door, locking it behind her, and finally leave through the exit.
 - Can you find other ways of efficiently locking all the doors?

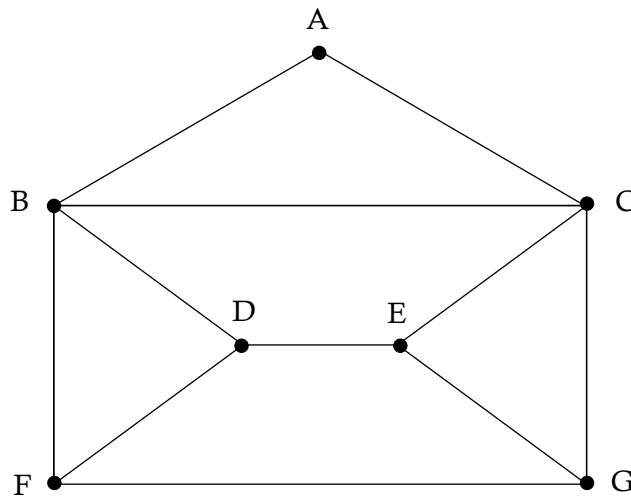


5. Gerry wants to draw the design shown below. What is the minimum number of times he will need to lift his pen off of the page to draw it (not including the time he lifts his pen once the design is drawn) if he doesn't want to go over any edge twice?

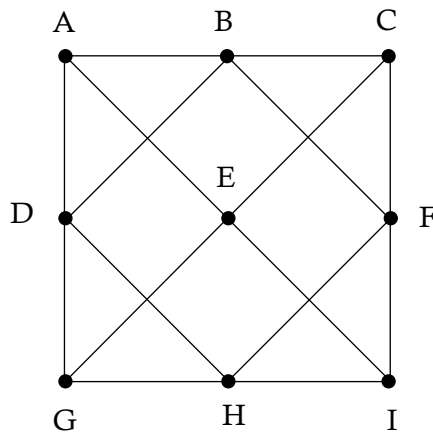


6. What is the maximum number of edges that can be in a trail in the graphs pictured below. (Bonus: can you justify your answer)

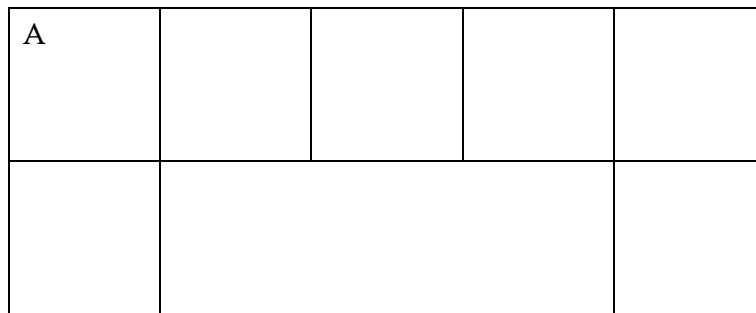
a.



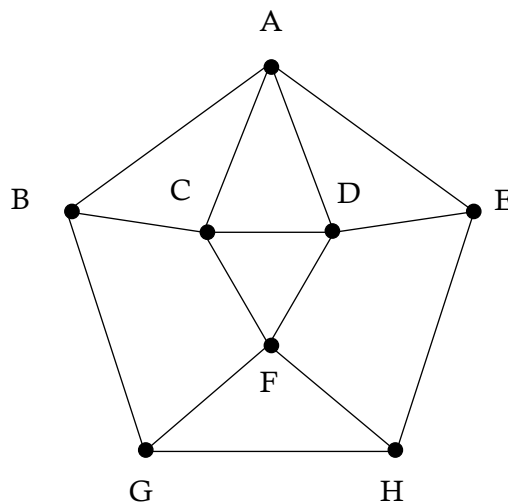
b.



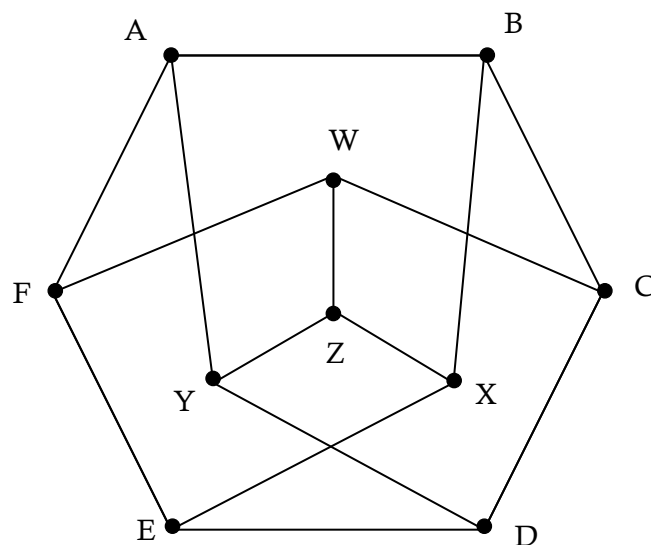
7. The diagram below shows the corridors of a building. There are various rooms between and around the corridors that aren't shown in the diagram. Each section of corridor is the same length. Each day a cleaner vacuums the entire length of the corridors, starting and finishing at a supply cupboard close to the vertex marked A.



- a. Explain why the cleaner will have to walk along some corridors more than once.
 - b. What is the minimum number 'repeated' sections of corridor that will have to be walked?
8. Determine whether the following graphs contain a Hamiltonian path. If they do, then is there also a Hamiltonian cycle? Describe any paths using the notation A-B-C-D (and so on), and prove any assertions you make.
- a.



- b.



9. For each of the graphs below determine the number of different Hamiltonian cycles in the graph.

For the purposes of this question, we will say that two cycles are the same if one is the reverse of the other, or if the relative order of the vertices is the same, but with a different start and finish point. For example, A-B-C-D-E-A and D-C-B-A-E-D are the same cycle.

